

Property (Ω) for Compacta on Finitely Many Parallel Lines

A higher-dimensional partial result for Conjecture 2.3 of arXiv:1611.06808

Automated proof attempt
agent_lane_03 model: GPT5.6

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Status: partial result, likely valid, pending human verification. The source proves (Ω) for vanishing ideals on the real line and conjectures it for arbitrary compact subsets of $\mathbb{R}^d$. Here the conjecture is proved in every dimension for arbitrary compact sets supported on finitely many parallel affine lines.

Source question

The excerpt is Conjecture 2.3 on printed page 6 of the source paper.

We do not know any compact set $K \subseteq \mathbb{R}^d$ such that the vanishing ideal $\mathcal{I}_K = \{f \in C^\infty(\mathbb{R}^d) : f|_K = 0\}$ does not satisfy (Ω) . To be concrete, we thus state a very optimistic conjecture:

Conjecture 2.3. *Every vanishing ideal $\mathcal{I}_K$ in $C^\infty(\mathbb{R}^d)$ satisfies (Ω) .*

For a compact $K \subset \mathbb{R}^d$, let

$$\mathcal{I}_K = \{f \in C^\infty(\mathbb{R}^d) : f|_K = 0\}.$$

The source proves that every closed ideal of $C^\infty(\mathbb{R})$ has property (Ω) [1, Theorem 2.1]. The following transfers that theorem to a genuinely higher-dimensional, geometrically broad class.

Theorem 1 (Parallel-line case of Conjecture 2.3). *Let $d \geq 2$, let $v \in \mathbb{R}^d$ be a unit vector, and let $b_1, \dots, b_q \in v^\perp$ be distinct. Put*

$$M = \bigcup_{i=1}^q (b_i + \mathbb{R}v).$$

For every compact set $K \subset M$, with no regularity or density assumption on its line slices, the vanishing ideal $\mathcal{I}_K$ in $C^\infty(\mathbb{R}^d)$ has property (Ω) .

Proof intuition

Restriction to the carrier M has a very explicit continuous linear right inverse: interpolate only in the transverse variable. This splits all smooth functions into a function vanishing on every carrier line plus independently chosen smooth functions along the lines. Imposing zero values on K then acts only on those one-dimensional components. The ambient ideal therefore becomes a topological direct sum of a complemented smooth-function subspace and finitely many one-dimensional vanishing ideals. Property (Ω) is stable under exactly these operations.

Proof

We first record the two standard permanence inputs, including their short proofs so that the argument does not depend on an unstated functional-analytic fact.

Lemma 2. *The Fréchet space $C^\infty(\mathbb{R}^d)$ has (Ω) . A complemented subspace of a Fréchet space with (Ω) has (Ω) , and a finite direct sum of spaces with (Ω) has (Ω) .*

Proof. Use the seminorms

$$p_n(f) = \max_{|\alpha| \leq n, |x| \leq n} |\partial^\alpha f(x)|.$$

Fix n , choose a smooth cutoff ψ equal to one on a neighborhood of the closed ball of radius n and supported in the ball of radius $m = n + 1$, and let ρ_δ be a standard mollifier at scale $0 < \delta < \delta_0$, with δ_0 small enough for the chosen neighborhood. If $p_m(f) \leq 1$, put $y = \rho_\delta * (\psi f)$. Taylor's formula and the convolution estimates give, for every $k \geq m$,

$$p_n(f - y) \leq C_n \delta, \quad p_k(y) \leq C_{n,k} \delta^{-k}.$$

Taking δ proportional to ε for $0 < \varepsilon < 1$, and $y = 0$ for $\varepsilon \geq 1$, is precisely the approximation formulation of (Ω) recalled in the source immediately before its Theorem 2.1.

If $P : X \rightarrow Y$ is a continuous projection, apply an (Ω) -decomposition in X to an element of Y and then apply P to both summands; continuity merely shifts the seminorm indices and constants. This proves the complemented-subspace assertion. Applying the defining estimates in each coordinate, taking the maximum of the finitely many indices and constants, proves the finite-direct-sum assertion. $\square$

Write every $x \in \mathbb{R}^d$ uniquely as $x = tv + y$, where $t = \langle x, v \rangle$ and $y \in v^\perp$. Choose $\chi_1, \dots, \chi_q \in C^\infty(v^\perp)$ such that

$$\chi_i(b_j) = \delta_{ij}.$$

For example, disjoint transverse bump functions suffice. Define

$$R : C^\infty(\mathbb{R}^d) \longrightarrow \bigoplus_{i=1}^q C^\infty(\mathbb{R}), \quad Rf = (t \mapsto f(tv + b_i))_{i=1}^q,$$

and

$$E : \bigoplus_{i=1}^q C^\infty(\mathbb{R}) \longrightarrow C^\infty(\mathbb{R}^d), \quad E(g_1, \dots, g_q)(tv + y) = \sum_{i=1}^q \chi_i(y) g_i(t).$$

Both maps are continuous for the usual smooth compact-open Fréchet topologies, and $RE = I$. Hence $P = ER$ is a continuous projection, $\ker P = \ker R = \mathcal{I}_M$, and

$$C^\infty(\mathbb{R}^d) = \mathcal{I}_M \oplus E\left(\bigoplus_{i=1}^q C^\infty(\mathbb{R})\right) \quad (1)$$

as a topological direct sum.

For each line define the compact slice

$$K_i = \{t \in \mathbb{R} : tv + b_i \in K\}, \quad J_i = \{g \in C^\infty(\mathbb{R}) : g|_{K_i} = 0\}.$$

Take the unique decomposition $f = h + E(g_1, \dots, g_q)$ from (1). Since h vanishes on every line and $E(g)|_{b_i + \mathbb{R}v} = g_i$, we have

$$f|_K = 0 \iff g_i|_{K_i} = 0 \text{ for every } i.$$

Consequently,

$$\mathcal{I}_K = \mathcal{I}_M \oplus E\left(\bigoplus_{i=1}^q J_i\right) \quad (2)$$

topologically.

It remains only to read (2). By Lemma 2, the complemented subspace $\mathcal{I}_M$ of $C^\infty(\mathbb{R}^d)$ has (Ω) . Each J_i is a closed ideal of $C^\infty(\mathbb{R})$, so it has (Ω) by the source theorem [1, Theorem 2.1]. Property (Ω) is invariant under topological isomorphism. The finite-sum part of Lemma 2 therefore shows that both summands in (2), and hence $\mathcal{I}_K$, have (Ω) . $\square$

Why this is only partial

The proof works whenever the compact set lies in a carrier whose restriction map has a continuous linear section and whose intrinsic vanishing ideal is known to have (Ω) . Arbitrary compact subsets of $\mathbb{R}^d$ need not admit such a one-dimensional carrier. A direct neighborhood cutoff does not close the gap: zero values on K do not control normal derivatives, so cutting a function off near K can fail even in C^1 .

The source's separate question about stability of the smooth extension property under unions also remains open. A naive fiber-product proof would require a controlled decomposition of functions vanishing on $K \cap L$ into $\mathcal{I}_K + \mathcal{I}_L$, and that algebraic equality can fail at accumulating intersections. The durable attempt note records eight distinct full-upgrade routes and their obstructions.

Reproducible check and literature boundary

The script `code/verify_parallel_line_splitting.py` constructs exact transverse Lagrange functions for three lines, verifies $RE = I$, and checks the restriction identities on 303 samples. This tests the explicit splitting, not the abstract (Ω) stability facts.

No duplicate was found in the run indexes. Exact-phrases searches and the later citations of the published source were checked through 11 August 2026. The only later citation returned by Semantic Scholar was a 2025 survey of Whitney-jet spaces, which does not settle Conjecture 2.3. This is a bounded novelty check, not an exhaustive priority claim.

References

- [1] L. Frerick, E. Jordá, and J. Wengenroth, *Extension operators for smooth functions on compact subsets of the reals*, Math. Z. 295 (2020), 1537–1552; arXiv:1611.06808. Included as `source_paper.pdf`.
- [2] D. Vogt, *Splitting of exact sequences of Fréchet spaces in the absence of continuous norms*, J. Math. Anal. Appl. 297 (2004), 812–832. Included under `references/` for the surrounding invariant and splitting framework; the permanence facts used above are proved directly.